

1. Which best describes the examinations required for ACCP Level II certification?

- a. General and Specific
- b. General and Practical
- c. General, Specific and Practical
- d. General, Specific, Practical and Instruction

CP-1, 7.1, Rev 8; 7.1.3 (last sentence)

2. In addition to the ASNT Basic and Method exams, what exams are required to become an ACCP NDT Level III certificate holder in a method?

- a. Specific and Procedure preparation exams
- b. Specific and Hands-On Practical exams
- c. Procedure Preparation and Practical exams
- d. Practical exam

ASNT Program Information Document CP-1, 7.2, rev 8

3. A monometer reading of a positive 4 inch of water on the side open to the atmosphere. What does this reading represent:

- a. 8 inch of water pulling a vacuum
- b. 8 inch of water of positive pressure
- c. 4 inch of water pulling a vacuum
- d. 4 inch of water of positive pressure

Pressure = density*h*gASNT NDT Level III

4. The document used to describe the ACCP program is:

- a. SNT-TC-1a
- b. CP-1
- c. CP-189
- d. ASME Section V

5. Reference standards containing artificial or natural discontinuities are generally required for accepting parts of the same configuration when using:

- a. PT
- b. ET
- c. RT
- d. MT

Not in M&P or Vol 10

6. Which of the following NDT methods may be used to test for bonding in a braze joint:

- a. Leak test
- b. Liquid penetrant testing
- c. Ultrasonic testing
- d. Visual testing

M&P 202, 2nd

7. A common tracer gas used because it is inert, non-toxic, a small molecule and inexpensive is _____.

- a. Hydrogen
- b. Oxygen
- c. Nitrogen
- d. Helium

Vol 10, 169, left, 3rd and 4th

8. Metal deterioration where relative vibratory movement between contacting surfaces occurs is _____.

- a. Crevice corrosion
- b. Fretting
- c. Fatigue
- d. Creep

M&P, 53, 4th

9. During application of a hydrophilic emulsifier by spraying it is found that the removal of surface penetrant is occurring too rapidly to control it effectively. What should be done to improved control of the removal of excess penetrant?

- a. Dilute the emulsifier with water
- b. Increase penetrant concentration
- c. Increase emulsifier concentration
- d. Increase emulsifier dwell time

~Vol 10. 113 rt and 115, left; no M&P

10. Which weld joint is best suited for radiographic examination?

- a. Lap
- b. Butt
- c. Edge
- d. Tee

~Vol 10, 236, Fig 48, left; no M&P

11. In visual testing, the SI unit for wavelength is _____.

- a. Nanometer
- b. Angstrom
- c. Foot-candle
- d. Lux

M&P 296, 1st; Vol 10, 21, rt, last; 49, left, 1st

12. The largest family of stainless steels is:

- a. Ferritic
- b. Austenitic
- c. Martensitic
- d. Pearlitic

Not in M&P

13. Eddy current techniques typically use frequencies in the range of ____.

- a. 0-10 Hz
- b. 50 to 1000 Hz
- c. 100 Hz to 10 MHz
- d. Over 100 MHz

M&P 346, 3rd; Vol 10, 258, rt, 1st

14. Rough surfaces can cause undesirable effects when performing an ultrasonic test. One such effect is ____.

- a. A change in material velocity
- b. The need for a less viscous couplant
- c. Focusing the sound beam
- d. Increasing the width of the front surface echo and consequent loss of resolution

Vol 10, 331, rt, 2nd

15. A radiographic exposure is made using a 4 minute exposure and a 24 inch source to film distance. If the SFD is changed to 12 inches and all other parameters are the same, what exposure time would be needed to achieve the same film density?

- a. 1 minute
- b. 2 minutes
- c. 8 minutes
- d. 16 minutes

No vol 10 or M&P

16. Brazing is a process using a filler metal with a melting point ____.

- a. Below 425 C (800 °F)
- b. Above 425 C (800 °F)
- c. Between 425 C (800 °F) and below 800 C (1500 °F)
- d. Of 100 C (212 °F)

M&P 201, 6th

17. A titanium fan disk from an aircraft engine has been removed from service. What NDT inspections would be preferred to detect fatigue cracks?

- a. Fluorescent penetrant and radiographic testing
- b. Radiographic and ultrasonic testing
- c. Eddy current and fluorescent penetrant testing
- d. Eddy current and ultrasonic testing

18. Which of the following NDT methods would be best for identifying weld cracks on the inside diameter of a large vessel containing a hazardous liquid?

- a. Liquid penetrant testing
- b. Thermal/infrared testing
- c. Ultrasonic testing
- d. Visual testing

19. Hall effect elements used in MFL testing units are: _____

- a. Very thick, solid state sensors
- b. Very thin, not more than 100 μm
- c. Relatively thick, solid state elements
- d. Similar to foerster microprobes in design and construction

Vol 10, 285, rt, last

20. Which of the following qualification levels is not addressed in CP-189?

- a. ASNT NDT Level III
- b. ASNT PdM Level III
- c. ACCP Professional Level III
- d. ASNT Responsible Level III

CP-189, para 2.2.1

21. The most commonly used UT technique for discontinuity discrimination is:

- a. Amplitude gating technique
- b. 6 dB drop technique
- c. Loss of back reflection monitoring
- d. B-scan image analysis

Vol 10, page 316, left, last; , 326, left, 3rd, 332, rt, 1st

22. Which type of discontinuity is usually detected best with wet magnetic particles?

- a. Slag inclusions
- b. Fatigue cracks
- c. Porosity
- d. Lack of fusion

Vol 10, left 299, surface; 306, rt, 1st

23. What will be the geometric unsharpness on a radiograph made using a 0.129 inch focal spot, 14 inch source to object distance, and 1.625 inch distance from the top of the part to the film?

- a. 0.017 inches
- b. 0.015 inches
- c. 0.160 inches
- d. 0.170 inches

Vol 10, 216

24. Which of the following leak test tracer gases is an ozone depleting CFC?

- a. Helium
- b. Sulphur hexafluoride
- c. Freon R-12
- d. Nitrous oxide

Vol 10, 158, left (lists chemicals but does not call them CFCs); M&P no help

25. Which of the neutron sources below is most commonly used for neutron radiography?

- a. Americium 242
- b. Californium 252
- c. Plutonium/beryllium
- d. Polonium 210

Nothing in MP and vol 10

26. Pipe threads on the end should normally visually tested using a _____ gauge.

- a. Cambridge
- b. Hi-Lo
- c. Pit depth
- d. Profile

Vol 9, 228, left

27. Test technique is defined in _____.

- a. The employer's certification process
- b. CP-189
- c. CP-105
- d. SNT-TC-1A

SNT-TC-1a, 5.3; CP-189, 2.1.24

28. Ideally a sampling plan should _____.

- a. Assume 85% confidence level on 90% of the parts
- b. Represent the smallest sample size possible
- c. Be selected on a non-random basis for 50% of the production items
- d. Accept all good lots and reject all the bad ones

M&P 287 no help; not in vol 10

29. In the review of 5 year old archived radiographs, discoloration and fading of the images is noted. This occurs as a result of ____.

- a. Failure to develop film at the proper temperature
- b. Leaving the film in the fixer more than twice the clearing time
- c. Failure to wash the film sufficiently
- d. Failure to dry the film sufficiently

Not in M&P or Vol 10

30. A large object may be radiographed using an array of film sheets placed side-by-side using a _____ collimator.

- a. Vertical
- b. Tangential
- c. Multiple slit
- d. Divergent (Disk)

31. Corrosion rates are dependent on several factors. With respect to the electrolyte present, which of the following is most correct?

- a. High concentration usually increases corrosion, the lower the acid concentration the higher the corrosion rate, and increased temperature lowers corrosion rates
- b. High concentration usually increases corrosion, the higher the acid concentration the higher the corrosion rate, and increased temperature increases corrosion rate
- c. Lower concentration usually increases corrosion, the lower the acid concentration the higher the corrosion rate, and increased temperature lowers corrosion rate
- d. High concentration usually increases corrosion, the lower the acid concentration the higher the corrosion rate, and increased temperature increases corrosion rate

Vol 3, 300, Etching Time; ~M&P 51-53

32. Which of the following penetrants would normally be the best choice for use on a part with rough surfaces and limited need for sensitivity?

- a. Color contrast, post-emulsifiable
- b. Fluorescent, water washable
- c. Fluorescent, post-emulsifiable lipophilic
- d. Color contrast, water washable

No m&p; no vol 10

33. Alloys subjected to cyclic tensile loading will most likely exhibit which type of discontinuity?

- a. Corrosion
- b. Seams
- c. Laps
- d. Fatigue cracks

M&P, 76, 2nd, Vol 10, 268, rt, 2nd, 3, rt, 3rd, 306, left, last; Vol 3, 206, left, 3rd

34. A possible hazard to consider when performing a leak test is:

- a. Exposure to radiation
- b. Flammability of the tracer gas
- c. Skin irritation from UV exposure
- d. Eye strain from arc flashes

Nothing in M&P or Vol 10; QA MS Level II #9, Level III #55

35. A radiograph of a weld has small, irregularly shaped areas of darker film density. Such indications are more likely associated with _____.

- a. GTAW
- b. PAW
- c. SAW
- d. Ultrasonic welding

Vol 10, 237-239 and ~M&P 214-221, 230

36. When leak testing using a hydrogen tracer gas, _____.

- a. Helium will collect in closed high stress areas
- b. Attention should be exercised to avoid inert gas asphyxiation
- c. Mixing of nitrogen and hydrogen should be avoided
- d. A nitrogen purge system is necessary

Nothing in either book M&P or Vol 10; Q&A Bub Level 1 #3C never use

37. Ultrasonic thickness measurement accuracy would be most affected by _____.

- a. Frequency
- b. Transducer diameter
- c. Couplant viscosity
- d. Material temperature

~Vol 10, 385, rt, 2nd

38. Which heat treat process restores carbon steel ductility following a drawing operation?

- a. Annealing
- b. Stress relieving
- c. Austempering
- d. Tempering

~M&P 58, last to 59. 1st; 182, 3rd; 49-50

39. A specimen supported on both ends with a notch in the middle struck by a pendulum opposite the notch is _____.

- a. Izod impact test
- b. Charpy impact test
- c. Tensile test
- d. Fatigue strength test

Nothing in M&P

40. A Hall effect probe is used to measure _____.

- a. Radiation level
- b. Flux density
- c. Impedance changes
- d. Acoustic emission rate

Vol10, 257, rt; 285, rt; 310, left, Fig 42; M&P 320, 1st; 354, 1st

41. A surface roughness measurement that is an arithmetic average deviation of peaks and valleys from a midpoint is _____.

- a. Average waviness value
- b. Root mean square value
- c. R_{max}
- d. Lay

Vol 9, 127 (rt, 3RD)-28; ~M&P 247

42. An increase in X-ray or gamma ray wavelength results in _____.

- a. The need for finer grain film
- b. Increased target heating
- c. Increased beam penetrating power
- d. Reduced beam penetrating power

M&P 340, 3rd; Vol 10, 200, rt, 4th

43. A dynamic casting process in which metal is forced against the outer mold surfaces to improve density is referred to as _____.

- a. Investment casting
- b. Continuous casting
- c. Centrifugal casting
- d. Permanent mold casting

M&P 158

44. A nonferrous metal that provides the combined properties of light weight and minimal cost is _____.

- a. Copper
- b. Magnesium
- c. Aluminum
- d. Nickel

M&P 101, 2nd; 99, 1st; 100, 1st; 92, 4th

45. Eddy current tests used to measure nonconductive coating thickness on a conductive substrate make use of _____.

- a. Conductivity
- b. Electromagnetic induction
- c. Lift-off
- d. Resistivity

Vol 10, 269, left, last

46. Acoustic emission testing for the purposes of structural health monitoring is capable of _____.

- a. Documenting event locations
- b. Predicting event location
- c. Relocating sensors to new locations
- d. Disabling and isolating bad sensors

Vol 10, 418, rt, 3rd; 423, rt, 4th and 5th

47. The SI unit of magnetic flux density is the _____.

- a. Tesla
- b. Weber
- c. Maxwell
- d. Gauss

Vol 10, 19, rt; 27, rt, 2nd last

48. A visual test technique utilizing stereo borescopes, video probes, and cameras to correct for distance or angle of view is _____.

- a. Direct visual testing
- b. Passive testing
- c. Remote (indirect) visual testing
- d. Active testing

Vol 10, 69-72; ~M&P 300-01

49. The detectability of a surface crack by magnetic particle examination is primarily a function of crack _____.

- a. Length
- b. Width, depth, and angle
- c. Surface texture and angle
- d. Depth

Subsurface: Vol 10, 280, left, 2nd; 313, 4th, M&P, 315, 1st

50. A rotating shaft subject to bending will likely exhibit _____ at a sharp change in cross section.

- a. Fatigue cracking
- b. Stress rupture failure
- c. Fracture mechanics
- d. Fretting corrosion

~M&P 76; 377

51. A powder metallurgy product involves compacting mixed metal powders and _____.

- a. Melting
- b. Sintering
- c. Forging
- d. Extrusion

M&P 192, 3rd and 4th

52. Compared to metals, acoustic emission probes used for fiber reinforced composites must be _____.

- a. Closer together
- b. Further apart
- c. Randomly spaced on one surface
- d. On both the OD and ID

Vol 10, 418, left, 3rd; ~338, left, 1st; ~420, rt, 2nd

53. Water chemistry, stress, and sensitization of a weld heat affected zone could lead to _____.

- a. Lack of fusion
- b. Stress corrosion cracking
- c. Enlarged porosity
- d. Tungsten inclusion induced stress cracking

M&P 90, 5th; 209, 1st & 5th; Sensitization Vol 9, 209, left, 3rd

54. Laminations in plate and sheet are normally oriented: _____.

- a. Normal to the plane of the material
- b. Parallel to the plane of the material
- c. Transverse to the rolling direction
- d. Angled to the plane of the material in proportion to rolling speed

Vol 10, 77, rt, 2nd; 303, rt, 4th

55. Standardization of test system sensitivity for ultrasonic straight beam is needed to _____.

- a. Amplitude of deeper flaws may be higher than those near the sound entry surface
- b. To ensure vertical trace is linear
- c. To allow comparison of indications with those from responses obtained from artificial discontinuities
- d. To compensate for poor back surface resolution

Vol 10, 316, rt, 4th and last; 326, left, 1st; 334

56. A liquid penetrant test of an aluminum fillet weld made with GMAW exhibits a linear indication at the toe. This is most probably _____.

- a. Incomplete penetration
- b. Slugging
- c. Overlap
- d. Slag inclusion

Vol 10, 78-83; 91-93

57. In a tempered forging quench cracks originate at the surface, but may extend entirely through a cross section. Which of the following would contribute most to the cause of a quench crack?

- a. Die pressure and cycle speed
- b. Surface defects and irregularities
- c. Uniform contours or cross section
- d. Proper tempering after quenching

Vol 10, 84, left, 2nd and 3rd

58. Conducting an extensive correlation study between NDT indications and destructive test results is considered _____.

- a. The best process for establishing accept/reject criteria
- b. Unnecessary as specifications and drawings for parts or surfaces under examination always specify the NDT test method or methods required for acceptance.
- c. Is unnecessary as instructions specify the accept/reject criteria, or the inspector relies on supplemental documents such as specifications
- d. Is unnecessary when liquid penetrant tests are performed on jet engine or nuclear components because no linear indications are acceptable

Vol 10, 125, left, 4th

59. The film density of a radiograph is dependent upon the _____.

- a. Radiation intensity and duration of exposure
- b. Radiation intensity
- c. Time and degree of agitation in stop bath
- d. Film characteristic curve

~Vol 10, 222

60. An aircraft wing composed of age hardenable aluminum alloy skin would exhibit abnormally high electrical conductivity in areas having _____.

- a. Low hardness
- b. High hardness
- c. High alloy content
- d. Uniform distribution of alloying elements

Vol 10, 264

61. Cold shuts, pipe, hot tears, and blow holes may be found in _____.

- a. Weldments
- b. Forgings
- c. Castings
- d. Machined components

M&P 164; Vol 9, 262, left; 182-83

62. For reliable penetrant testing personnel should be _____.

- a. Trained and experienced in the processing steps
- b. Certified Level II
- c. Independent and work for a third party organization
- d. Certified Level III

~Vol 10, 16-17

63. Other factors being the same, radiographic contrast is improved by _____.

- a. Higher KeV and lower mA
- b. Adjusting exposure to minimize film density to the least necessary
- c. Lower KeV
- d. Adjusting KeV to produce minimum film density required

Vol 10, 220, rt, last

64. What test is used to monitor hydrophilic emulsifier concentration?

- a. Flash point
- b. Wetting ability
- c. Fluorescence
- d. Refractometer
- e. Viscometer

Vol 2, 4th ed, 132, left last; 510, left, 3rd

65. The three most common crystal structures (unit cells) in metals are:

- a. DCP, BCC, FCC
- b. HCP, BCC, HFCC
- c. BCC, FCC, and Hexagonal Far Packed
- d. BCC, FCC, HCP

M&P 36-39

66. Which process makes high quality castings due to minimizing oxygen contact with the molten metal?

- a. Continuous casting
- b. Centrifugal casting
- c. Investment casting
- d. Permanent mold casting

Not in Vol 10 or M&P

67. Which of the following is an important consideration of acoustic emission testing?

- a. Unstressed flaws will not produce emission
- b. Acoustic emission waves only travel short distances
- c. Acoustic emission signals are usually buried in noise
- d. There are no standard ways to evaluate the data

M&P 360, 2nd

68. Calibration of a UT system by adjusting sensitivity to a series of flat bottom holes at different depths to 50% of the screen height is done to obtain a:

- a. Resolution level
- b. Reference line
- c. Refracted angle
- d. Exit point

~Vol 10, 316, rt, 4th, 316 Table 1; 318 left

69. In an aluminum-boron composite, matrix bonding is achieved by _____.

- a. Explosion bonding
- b. Diffusion bonding
- c. Cladding
- d. Resin transfer molding

ASM Vol 21, p1373 (10th green ed); Nothing in Vol 10 or M&P

70. When determining fluid level in a tank with IR, the principle being used is thermal_____.

- a. Resistance
- b. Capacitance
- c. Diffusivity
- d. Effusivity

Vol 10, 185, #3; Snell and Fluke Articles; Not in M&P

71. What material property would allow eddy current to separate 300 from 400 series stainless steels?

- a. Magnetic permeability
- b. Liftoff or fill factor
- c. Conductivity
- d. Frequency

M&P 90, 6th and 7th; 314, 1st; 348, 3rd and Fig 12.34; Vol 10, 496, Fig 4; ~265, left, 3rd

72. Which of the following is the most sensitive leak test?

- a. Mass spectrometer, detector probe
- b. Mass spectrometer, envelop
- c. Halogen diode
- d. Bubble leak

Vol 10, 152, Table 3 (lowest number); 149, left, 2nd, Fig 7; 147, left; QA MS 3 #17

73. Loss of material due to slight relative, vibratory movement between two contacting surfaces is referred to as _____.

- a. Fretting
- b. Contact stress fatigue
- c. Corrosion
- d. Erosion Corrosion

M&P , 53,4th bullet; Vol 9, 206, left, 2nd

74. What effect would a rough entry surface have on an ultrasonic test?

- a. Improved ability to detect flaws near the sound entry surface
- b. Greater sensitivity
- c. Widening of the sound entry signal and loss of resolving power
- d. Reduced noise attributed to side lobe energy normally obtained from smooth surfaces

75. A Hall effect probe is used to _____.

- a. Locate studs hidden in walls
- b. Measure current density when performing eddy current testing
- c. Measure radiation leaking through the shielding of gamma ray isotope cameras
- d. Measure flux leakage during magnetic particle and magnetic flux leakage testing

76. Penetrant should not be used on which of the following?

- a. Porous aluminum
- b. Full density powder metallurgy parts
- c. Glazed ceramics
- d. Forged steel

77. What is not defined in CP-189?

- a. ACCP NDT Level 3
- b. ASNT NDT Level 3
- c. ASNT PdM Level 3

2.2.1

78. The most sensitive leak test technique is _____

- a. Helium mass spectrometer with a sniffer
- b. Helium mass spectrometer within a vacuum enclosure
- c. Bubble leak
- d. Halogen diode

Vol 10, 152, Table 3 (lowest number); 149, left, 2nd, Fig 7; 147, left; QA MS 3 #17

79. A thermal infrared technique to measure fluid level in a tank relies on _____.

- a. Thermal capacitance

Vol 10, 185, #3; Snell and Fluke Articles; Not in M&P

80. Ultrasonic standards having equal sound paths but reflectors of small to larger diameters are:

- a. Distance amplitude correction blocks
- b. Area amplitude blocks
- c. Used to correct for attenuation of sound in the material
- d. Used to correct poor horizontal linearity

81. How far apart are the threads if the pin advances 1/50 of an inch with each rotation?

- a. 50 threads per inch
- b. 50 inch per thread
- c. 0.02 threads per inch
- d. 0.02 inch

82. A written practice uses the minimum recommendations of SNT-TC-1A. Do the following NDT Level II examination scores meet the requirements of the written practice? General 98%, Specific 69.4%, and Practical 87%

- a. Yes, because the composite score is above 80%
- b. No, the minimum passing grade for a written exam is 70%
- c. No, the general examination score is below 70%
- d. No, the composite score is less than 80%

8.1.4 and 8.1.5

83. What would be the best magnetic particle test technique to use on a cylindrical part?

- a. Coil followed by head shot
- b. Cable wrap
- c. Head shot followed by coil shot
- d. Cable wrap and prods

84. Which of the following is most correct regarding the topics in CP-105 for Level I and II?

- a. The topics are progressive from Level I to II
- b. The topics for Level I are independent of those in a Level II class.
- c. A Level II class includes all the topics in Level I, plus additional topics
- d. Once the Level I and II classes are successfully completed, the attendee is certified

85. An ACCP NDT Level III candidate must demonstrate that they can write a procedure. To fulfill this requirement, the candidate _____.

- a. Must write a procedure to ASME Section V.
- b. Must accomplish a procedure preparation exam administered by ASNT
- c. May use a prior procedure written for a similar component and acceptance standard
- d. May use a previously drafted procedure accepted by the employer

86. Five half value layers of shielding would reduce 128 mR/hr to ____.

- a. 4 ci
- b. 25.6 mR/hr
- c. 4 mR/hr
- d. 12.8 mR/hr

87. A radiographic exposure takes 40 minutes at 24 inches. How long would it take to obtain the same degree of film darkening if the exposure was done at 12 inches?

- a. 80 minutes
- b. 20 minutes
- c. 160 minutes
- d. 10 minutes

88. Which ACCP exam is not required in SNT-TC-1a and CP-189 for Level II?

- a. General
- b. Specific
- c. Instruction
- d. Practical

SNT-TC-1a para 8.3.5, 8.4.4 and 8.5; CP-189 para 1.5 and 6.3

89. Which wave would be dampened out during immersion testing?

- a. Creeper
- b. Longitudinal
- c. Shear
- d. Surface

90. Which of the following is not addressed in SNT-TC-1a?

- a. Level II limited
- b. ACCP NDT Level II
- c. Trainee
- d. Instructor

SNT-TC-1a para 1.5, 2, 4.2, 6.3.1

Question	Answer	Question	Answer	Question	Answer
1 d		41 b		81 d	
2 c		42 d		82 b	
3 b		43 c		83 c	
4 b		44 b		84 a	
5 b		45 c		85 b	
6 c		46 a		86 c	
7 d		47 a		87 d	
8 b		48 c		88 c	
9 a		49 a		89 D	
10 b		50 a		90 D	
11 a		51 b			
12 b		52 a			
13 c		53 b			
14 d		54 b			
15 a		55 c			
16 b		56 c			
17 c		57 b			
18 c		58 a			
19 b		59 a			
20 d		60 a			
21 a		61 c			
22 b		62 a			
23 b		63 c			
24 c		64 d			
25 b		65 d			
26 d		66 c			
27 a		67 a			
28 d		68 b			
29 c		69 b			
30 d (Disk)		70 b			
31 b		71 a			
32 d		72 b			
33 d		73 a			
34 b		74 c			
35 c		75 d			
36 d		76 a			
37 d		77 None			
38 a		78 b			
39 b		79 A			
40 b		80 b			

